Historical Accuracy of Rolling Focus IPCA Forecasts

Abstract

- Main question: how do Focus-based IPCA forecast errors vary across rolling 3-, 6-, and 12-month horizons?
- Secondary questions:
  - How is 12-month forecaster disagreement related to error?
  - How does a quarterly refitted AR(1) forecasting agent compare with an agent using Focus forecasts on matched dates?
- Main finding: cumulative mean absolute error (MAE) is 0.434, 0.953, and 1.794 percentage points, respectively, across 279 matched forecast months.
- On the 73 quarterly origins used for the agent simulation, Focus has lower MAE at all three horizons; the 12-month paired bootstrap interval includes zero.

1 Introduction

- Motivate the comparison of past Focus expectations with realized Brazilian IPCA.
- Introduce the quarterly AR(1) forecasting agent inspired by Poledna et al. [1], and compare it with Focus on identical dates and horizons.

2 Methods

2.1 Data

The Central Bank of Brazil publishes a weekly suvey on the expectations of economic indicators from financial market players, called the Focus Report [2]. From this survey, we can recorver the estimate for the consumer inflation rate of Brazil, the IPCA, for following 3, 6 and 12 months. On the other hand, the Brazilian Institute for Statistcs (IBGE) publishes the monthly realized value for the IPCA [3]. This study will use data from both of these sources to calculate statistics and produce simulations regarding the error between expected and realized value of inflation in Brazil. We use 279 forecast dates, one per month from November 2001 through January 2025.

2.2 Forecasts and error measures

The core measurement for this work will be the signed error for the inflation expectations, defined as the median forecast minus the realized inflation. From this, we will also relport absolute-error measures, including mean absolute errors (MAE), quantiles, and empirical cumulative distribution functions (ECDFs). We will also explore if the signed error is related to the standart deviation of the prediction.

2.3 Annual target check: direction and closeness

We also analyse the closeness of the inflation forecasts, defined as $|\hat{\pi}_y - \pi_y|/|\pi_y|$, where $\hat{\pi}_y$ is forecast IPCA and π_y is realized IPCA. Call a forecast "90% close" when the error is at most 0.10 (within 90–110% of the realized value), and "95% close" when it is at most 0.05. This treats under- and overestimation symmetrically. We will calculate the signed-error distribution around zero and an ECDF of relative error with markers at 5% and 10%.

2.4 Simulation

In the macroeconomic agent based model of Poledna et al. [1], the agents use a fitted AR(1) rule for estimating the inflation for the next quarter. We will apply a similar rule in this study and compare it with the predictions from the focus report. The data will be 73 quarterly origins from January 2007 through January 2025.

3 Results

3.1 Main finding and descriptive statistics

Constructed horizon	MAE (p.p.)	Mean signed error (p.p.)
3 months	0.434	-0.165
6 months	0.953	-0.436
12 months	1.794	-1.033

- The 12-minus-3-month MAE difference is 1.360 p.p.; its 12-month-block bootstrap 95% interval is 0.788–2.089 p.p. The block-bootstrap approach follows Künsch [4].
- In the separate quarterly agent comparison, Focus MAE versus AR(1) MAE is 0.396 versus 0.648 p.p. at 3 months, 0.866 versus 1.160 p.p. at 6 months, and 1.572 versus 1.821 p.p. at 12 months.
- The quarterly paired-bootstrap intervals for Focus's MAE gain are 0.133–0.376, 0.076–0.495, and -0.224–0.692 p.p., respectively; the 12-month interval includes zero.

3.2 Published 12-month benchmark and disagreement

- Published 12-month MAE: 1.809 p.p.; mean absolute gap from the constructed 12-month forecast: 0.049 p.p.
- Spearman correlation between published forecast dispersion and subsequent absolute error: $\rho = 0.245$ (bootstrap 95% interval: 0.046–0.413).
- Show the forecast comparison, disagreement scatter plot, bootstrap distribution, and convergence diagnostics.

3.3 Annual target check: descriptive results

- Of 25 annual forecasts, 16 underestimated realized IPCA and 9 overestimated it; mean signed error was -0.984 p.p.
- Seven forecasts (28%) were within 10% of realized inflation; one (4%) was within 5%. Show these counts on the relative-error ECDF.

4 Discussion

- Interpret the 279-month Focus results while accounting for the different scales of cumulative 3-, 6-, and 12-month inflation.
- Compare the two forecasting agents within each horizon. The quarterly AR(1) rule uses lagged realized IPCA; Focus uses professional expectations. This is not a model of agent interaction or causal effects on inflation.
- Keep the annual 25-year check distinct from the 279-month rolling analysis. The 90% and 95% closeness thresholds are descriptive tolerances, not confidence levels.
- Discuss overlapping windows, possible macroeconomic regime changes, aggregate Focus statistics, the assumed IPCA release lag, and historical data revisions.
- Identify questions for further analysis.

References

[1] Poledna, S., Miess, M. G., Hommes, C., and Rabitsch, K. (2023). Economic forecasting with an agent-based model. *European Economic Review*, 151, 104306. `https://doi.org/10.1016/j.euroecorev.2022.104306`.

[2] Banco Central do Brasil. Expectativas de Mercado. Portal de Dados Abertos. `https://dadosabertos.bcb.gov.br/dataset/expectativas-mercado`.

[3] Instituto Brasileiro de Geografia e Estatística. *Índice Nacional de Preços ao Consumidor Amplo (IPCA)*, SIDRA table 1737. `https://sidra.ibge.gov.br/tabela/1737`.

[4] Künsch, H. R. (1989). The jackknife and the bootstrap for general stationary observations. *The Annals of Statistics*, 17(3), 1217–1241. `https://doi.org/10.1214/aos/1176347265`.

A Additional Methods and AI Use

- Add any detailed data checks and simulation diagnostics needed beyond the main article.
- Explain how AI was used in research, coding, and writing, and how the results were checked.

B Implementation

- Include key R code or point to the supplementary analysis notebooks.
- Point to `ipca_quarterly_agent_comparison.ipynb`; record the Focus snapshot, IBGE source, filters, package versions, and bootstrap seed.